

Life in tropical forest

Some tropical forest animals spend all their time on the ground. For most, however, daily life involves getting around among trees. The canopy holds most of the forest's food, so an animal that is good at moving around in the tree tops has the greatest chance of thriving. Some animals are so well adapted to life in the trees—breeding as well as feeding there—that they very rarely have to visit the forest floor.

Moving in trees

Small animals need relatively few special adaptations for moving about in trees. Ants, for example, are so light that it makes little difference to them whether they are traveling up and down trees or across the forest floor. But for larger animals, such as apes, monkeys, and other primates, climbing is a dangerous occupation: if they lose their grip—as occasionally happens—they risk a fatal fall.

Most primates climb by running or leaping along the tops of branches, often using familiar routes that act like highways through the trees. Monkeys follow these routes mainly by sight, but many of the more primitive primates, such as bushbabies, move around after dark, identifying their pathways partly by smell. Gibbons are different again: they



PREHENSILE TAILS

Climbing snakes—such as this emerald tree boa—use their tails to clamp themselves to branches. The front of the snake's body folds up, ready to straighten out as it strikes anything that comes within reach.

travel underneath the branches by swinging hand-over-hand in a breathtakingly acrobatic manner. This unusual but highly effective form of movement is called brachiation.

Tropical forest harbors a huge variety of flying animals—birds, bats, and flying insects—that swoop or hover among the tree tops. However, during the course of evolution, many unrelated animals, including mammals, frogs, and even snakes, have developed winglike flaps of skin that enable them to glide. Some of these gliders can travel over 330 ft (100 m) from tree to tree and, remarkably, many of them are most active after dark.

Communication

In any kind of forest, animals face problems keeping in touch. In the canopy, leaves and branches make it difficult to see for more than a few yards, while tree trunks get in the way on the ground. As a result, many forest animals rely on sound and scent, rather than visual signals, to claim territories and attract partners. Some of the loudest animals in the world live in tropical forest. They include howler

monkeys, bellbirds, parrots, cicadas, and an enormous variety of tree frogs. Like



PRIVATE CALL

With its vocal sac inflated, a male tree frog calls to deter rival males and to attract potential mates. Female frogs react only to the call sign of their own species; and the louder and longer a male calls, the more likely females are to respond.

mammals and birds, each species of tree frog has its own characteristic call: some produce a short metallic “tink”; others generate a sustained trilling that sounds like machinery.

Signaling with sound can be dangerous because it can attract predators as well as potential mates. Tree frogs and cicadas minimize the problem by pitching their calls so that the source is very difficult to locate. Other animals, including many mammals and flying insects, avoid the problem by using scent to stay in touch. One great advantage of scent is that it lingers: for example, in marking its territory, a jaguar or okapi leaves a signal that will last for several days.

Keeping out of danger

Tropical forest abounds with camouflaged animals as well as species that mimic others. Animals that use camouflage—chiefly insects and spiders, but also snakes, lizards, frogs, and toads—resemble a huge variety of inanimate objects, from bark, thorns, and bird droppings to branches and fallen leaves. Many animals use camouflage to avoid being spotted and eaten, but some predators also use it to enable them to ambush their prey.

Mimicry, in which one species “pretends” to be another, is a subtler means of avoiding attack. It involves a relatively harmless species evolving to look like one that is dangerous, and it is most common in invertebrates.

Some tropical forest spiders, for example, closely resemble stinging ants and even move like them. Matters are complicated where several species come to look alike. Some groups of unrelated butterflies, which contain poisons that are distasteful to birds, imitate each other; thus they have evolved the insect equivalent of a shared warning trademark.

Warning signals are most developed in extremely toxic animals. For example, unlike other frogs, tiny poison dart frogs hop nonchalantly about the forest floor, relying on their extraordinarily vivid colors to warn other animals that they are not merely unpalatable but highly dangerous to eat.



ARM OVER ARM

Swinging, or “brachiation,” is a highly efficient way of moving around in trees. This lar gibbon can leap over 16 ft (5 m) between hand holds and can easily overtake someone running on the ground.